Spatial Analysis

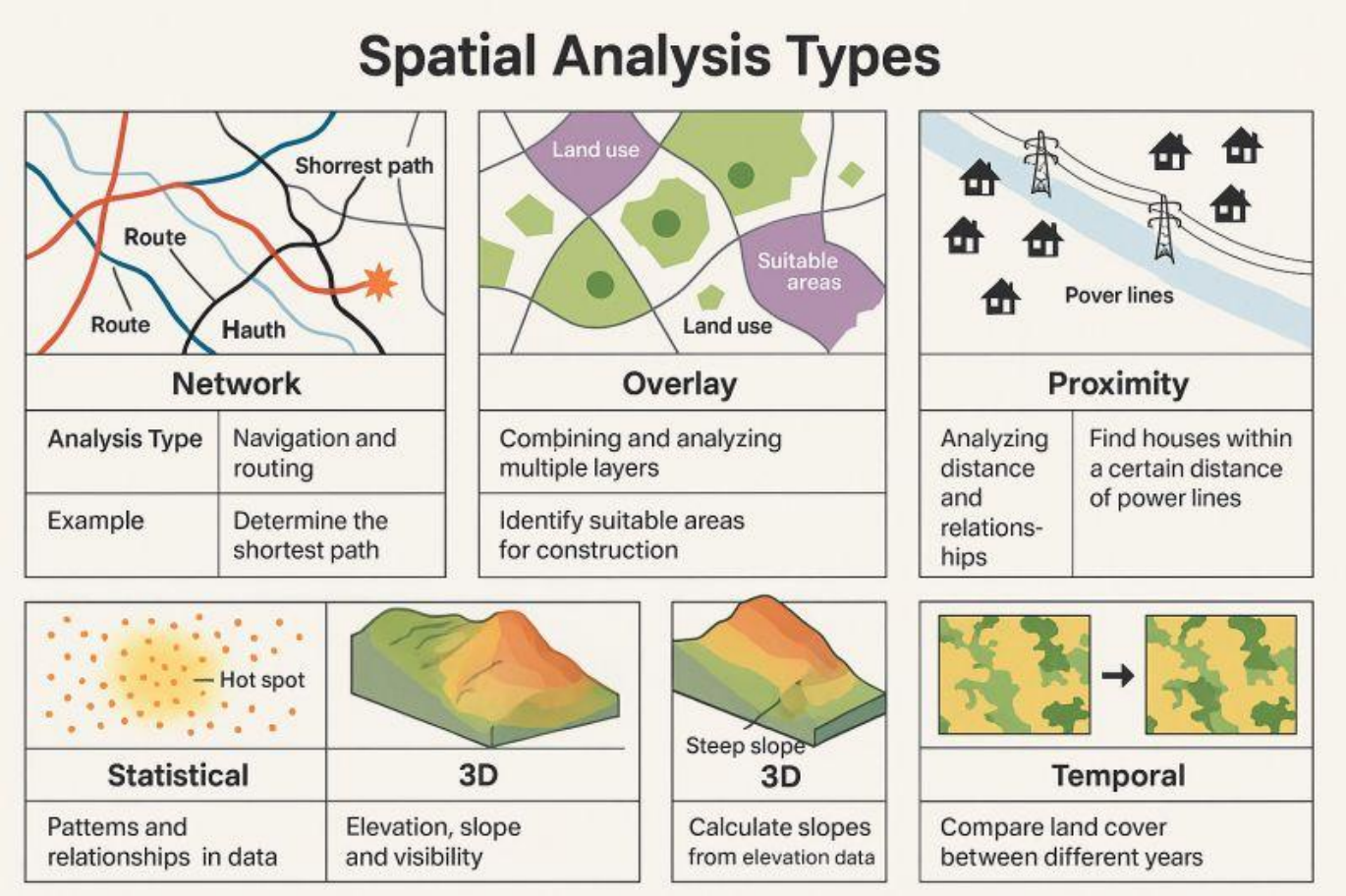
Lecture content

- Concepts of spatial analysis and modelling.
- Spatial query using attributes and locations.
- Spatial operations.
- Spatial interpolation methods and errors.

Spatial analysis definition and types

- A set of methods used to query features based on their attributes or analyse how layers interact spatially.
- Used to identify patterns, associations, and evidence of change through time and location.
- **Spatial query:** filtering and selecting data from a geodatabase based on attributes and location.
- **Spatial operations:** a type of spatial query class that includes operations such as intersection, containment, and distance calculations for objects in a given region or point.
- **Spatial interpolation:** estimates unknown values at specific locations based on known values from surrounding points.

Spatial analysis and modelling

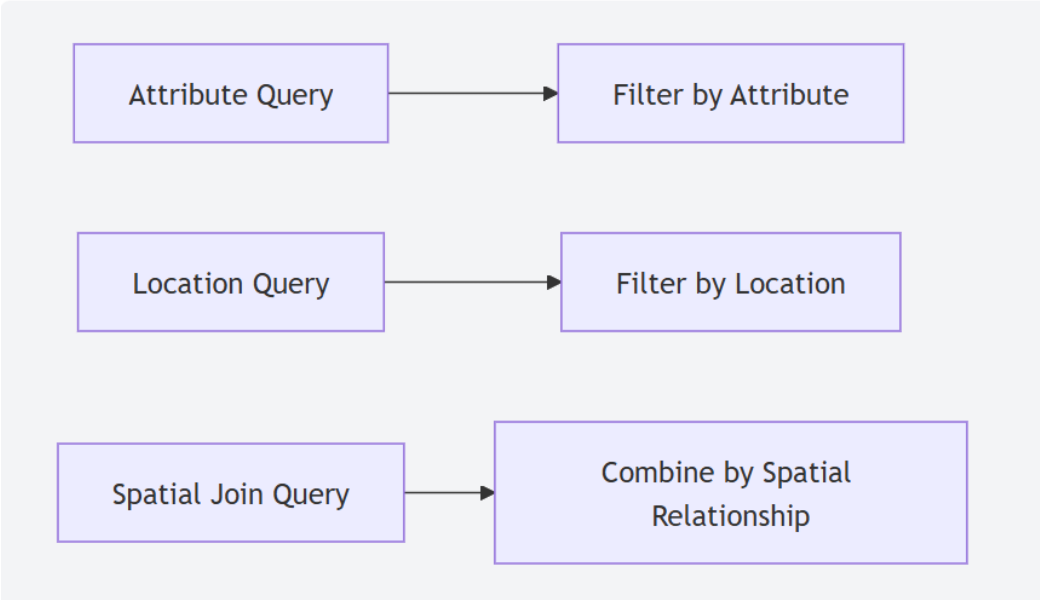


Spatial query

- Select, extract and interactive query.
- Query by attribute.
  - To retrieve all features with particular **attribute** values from one data layer or feature class in a spatial database and display their locations in a map.
- Query by location.
  - To find features from one or multiple data layers based on their **locations** relative to features from another layer in the same spatial database.
  - To identify **attribute** values associated with features at particular locations.

Query by attribute

- A query uses a specific database query language.
- Many GIS systems support SQL (Structured Query Language).



Query by attribute and location

Attribute: 'Retrieve all water mains installed before 2010 with a diameter less than 30cm'.

```
SELECT * FROM w_mains WHERE "year" < 2010 AND "diameter" < 30
```

The WHERE clause filters the results to only include those water mains that meet the two criteria.

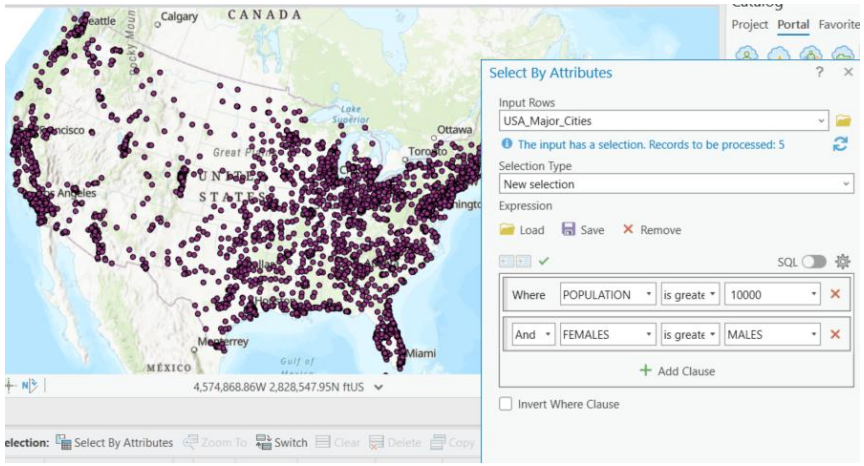
Location: 'Select all buildings that intersect with a given park'.

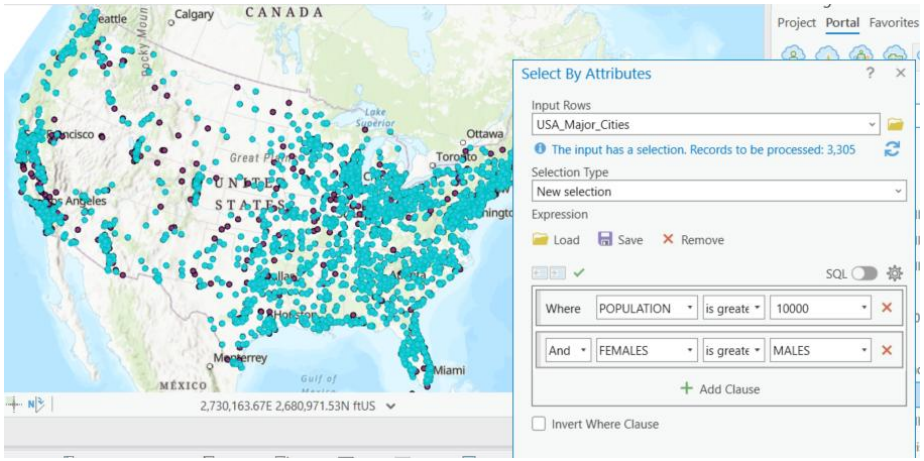
```
SELECT * FROM buildings WHERE ST_Intersects(buildings.geometry,
(SELECT geometry FROM parks WHERE name='Whitworth'))
```

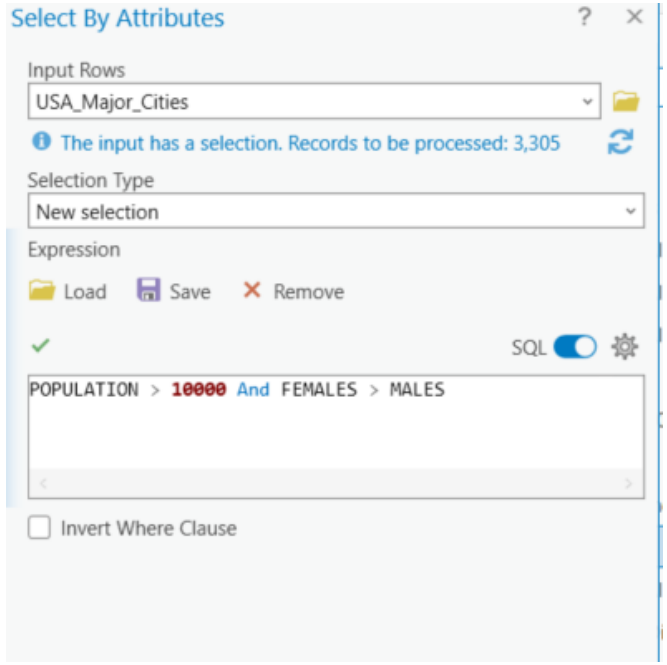
The ST_Intersects function checks if the geometry of each building in the dataset intersects with the geometry of Whitworth Park.

There are large number of ST_geometry functions that can be used in a GIS.

Example – query by attribute

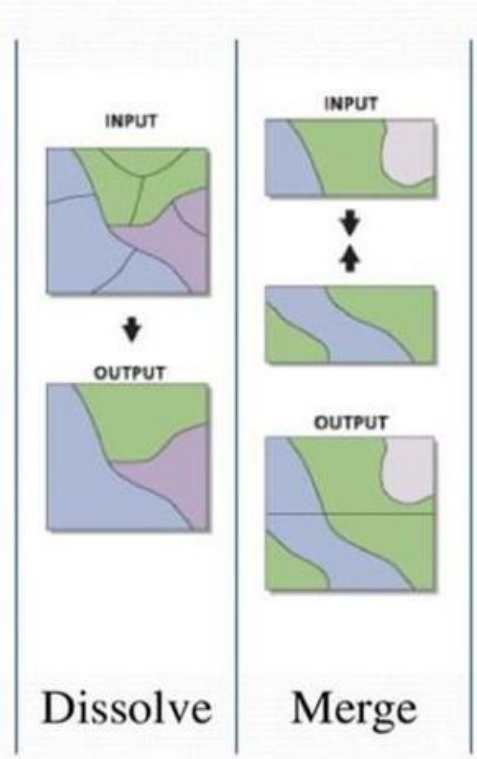


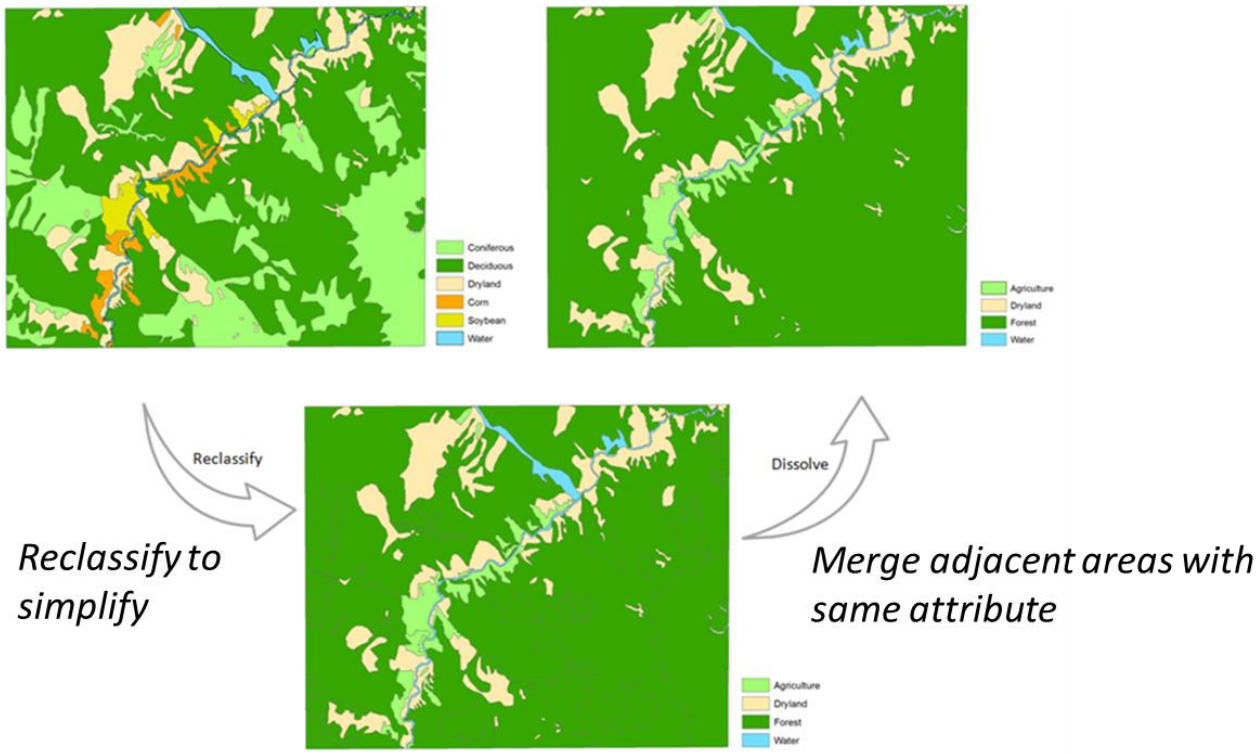




Geoprocessing operations (merge, reclassify and dissolve)

Assigning new values to raster dataset cells based on their pre-existing values is known as reclassification in GIS.





Overlay analysis

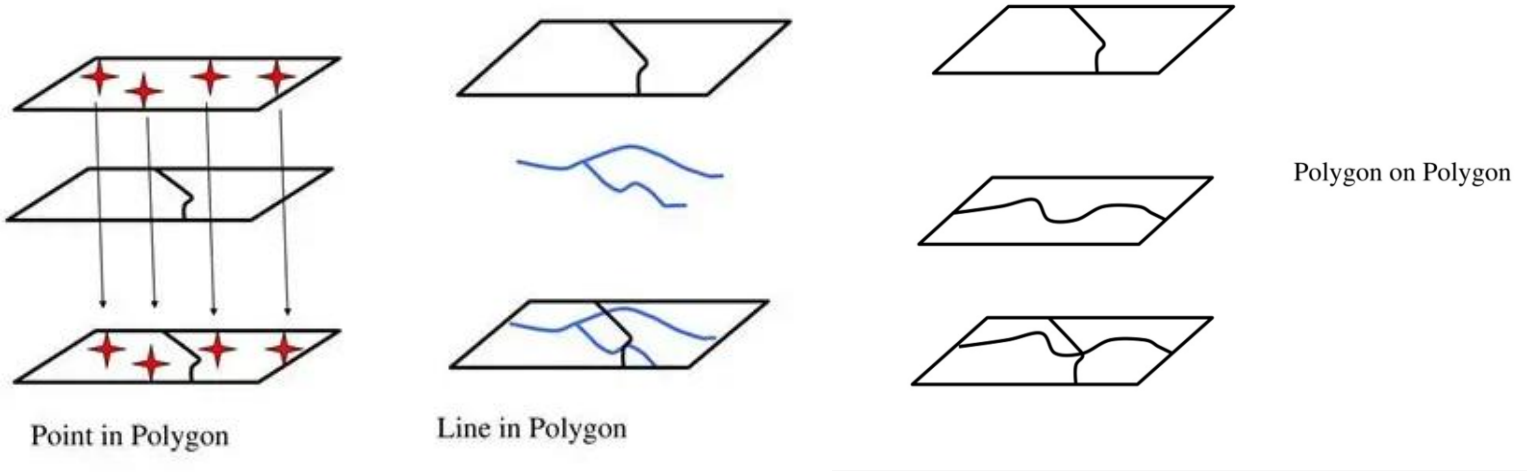
- A process of stacking multiple data layers with a common georeferencing system on top of each other so that the relationships between features at each location can be analysed.
- The most common GIS operation for comparing and analysing multiple data layers simultaneously.

  E.g. calculating the number of properties within a flood risk zone.
- Overlay analysis can be conducted using either vector or raster data.

Vector overlay - 1

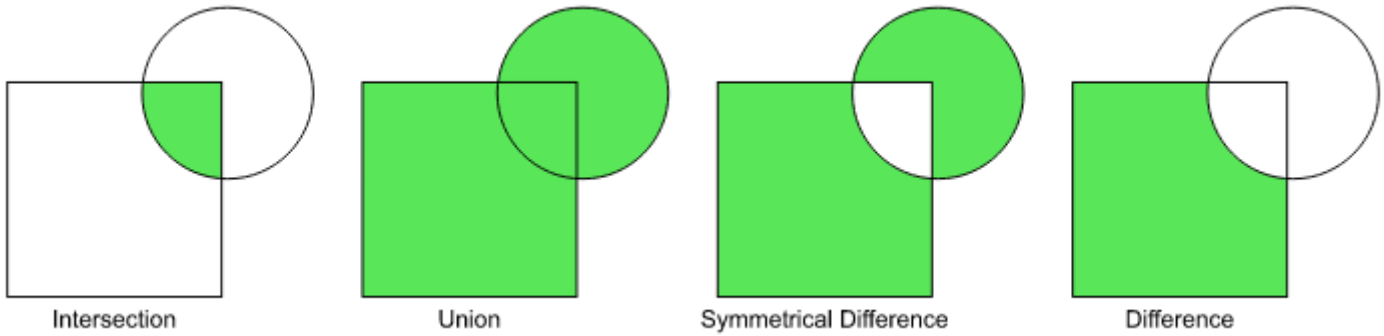
- Two or more vector layers are combined to produce new geometries.

e.g. union



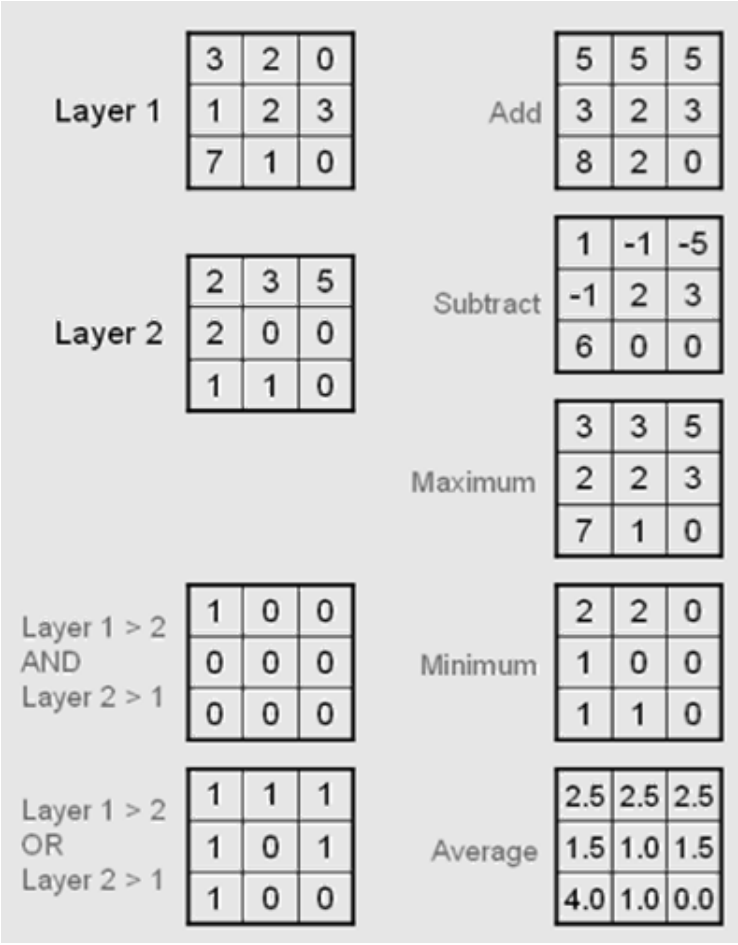
Vector overlay - 2

- Two or more vector layers are combined to produce new geometries.
- Typical overlay operations include *union*, *intersection*, and *difference* - named after the result of the combination of two layers.



Raster overlay (map algebra)

- Raster overlay superimposes at least two input raster layers to produce an output layer.
- Each cell in the output layer is calculated from the corresponding cells in the input layers.



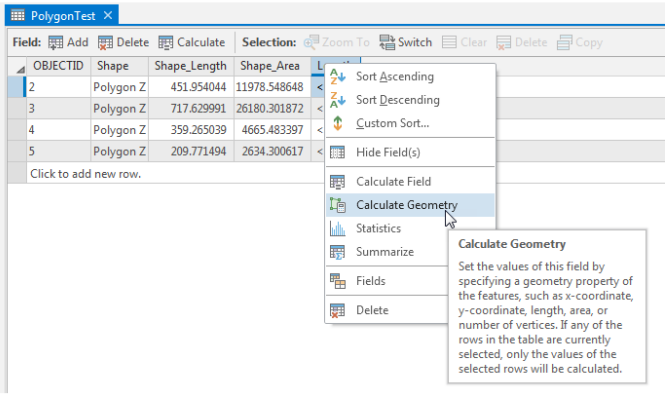
Map algebra

Type	Operators/functions
Arithmetic	+, –, ×, ⁄, square, square root, exponentials, logarithms
Logical	AND, OR, NOT
Relational	=, >, ≥, <, ≤
Conditional	IF…THEN…, IF…THEN…ELSE
Trigonometric	(inverse) cosine, (inverse) sine, (inverse) tangent
Statistical	maximum, minimum, mean, range, standard deviation, sum, median, majority, minority, variety

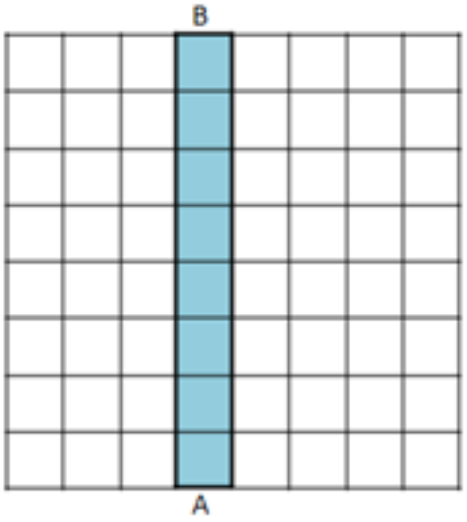
Geometric and distance measurement

- Length, area, perimeter and shape. Useful for landscape metrics e.g. interaction between spatial pattern and ecological processes.
- Geometric properties and distances are usually not encoded during spatial data input. GIS offers measurement functions to calculate them.

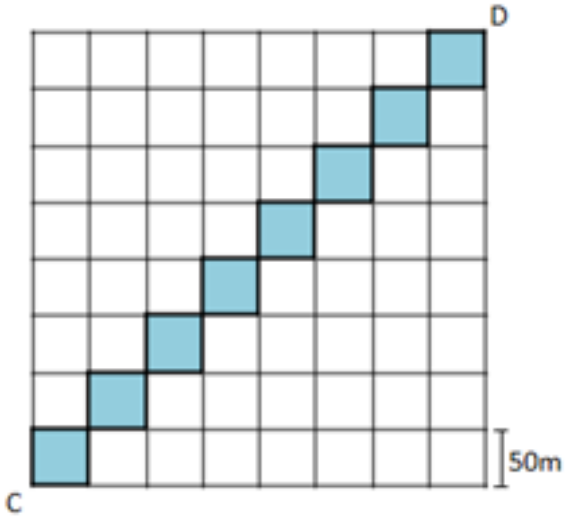




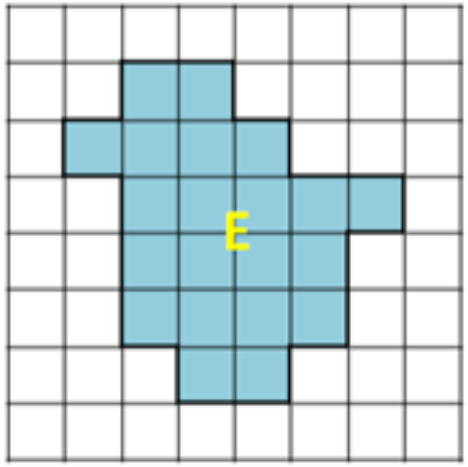
Measuring length, perimeter and area in raster



The length of line AB
= 8 cells × (50m / cell)
= 400m



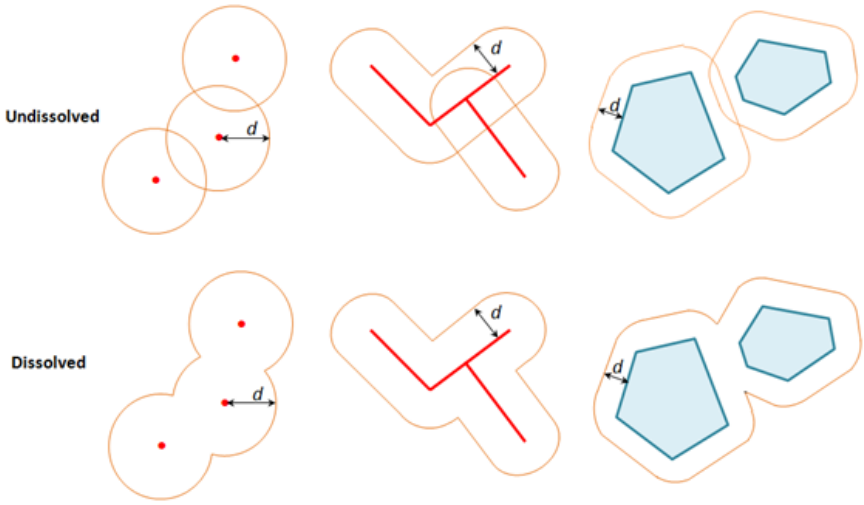
The length of line CD
= 8 cells × (50m × 1.414 / cell)
= 565.6m

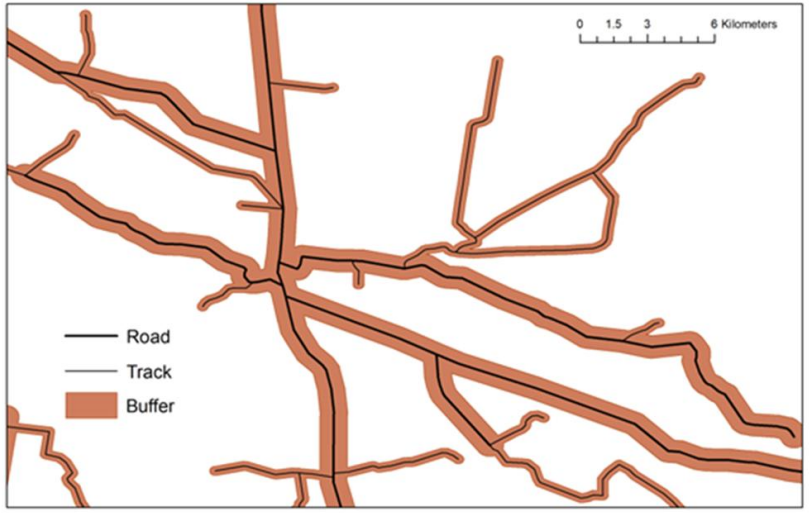


The area of polygon E
= 21 cells × (50m × 50m / cell) =
52500m^2

The perimeter of E
= (4 × 21 cells – 2 × 30 sides) × 50m
= 1200m

Neighbourhoods: Buffers





Buffering is a distance measurement tool for vector data commonly available in GIS. It creates a zone around a feature or a set of features with a specific width.

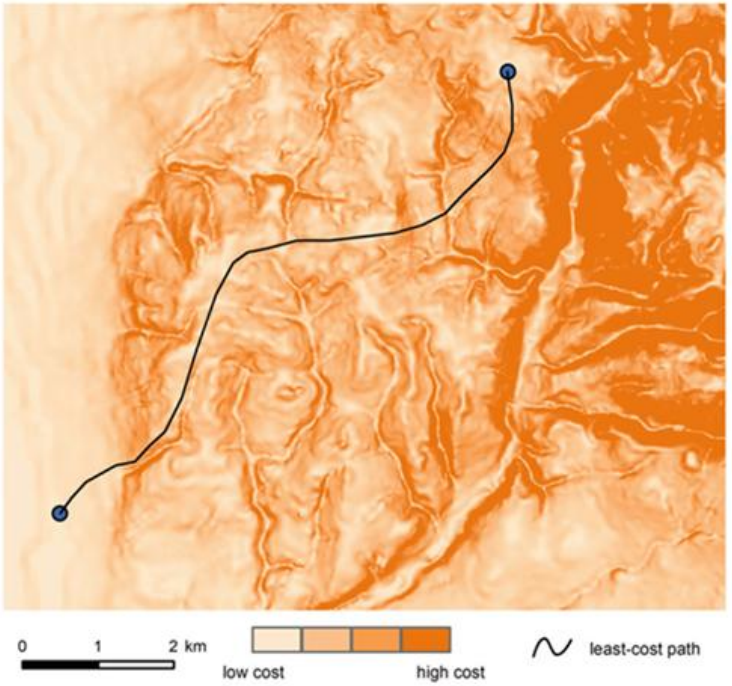
Neighbourhoods: Cost distance surface



- Cost-distance analysis measures how costly, or expensive, it is to travel over a surface.
- Simulates how a mover will experience the landscape as they travel through it. This is different from Euclidean distance, which captures how far a mover is from known locations in a straight line.
- Cost can be measured in topography, time, money, energy expended, or other forms.

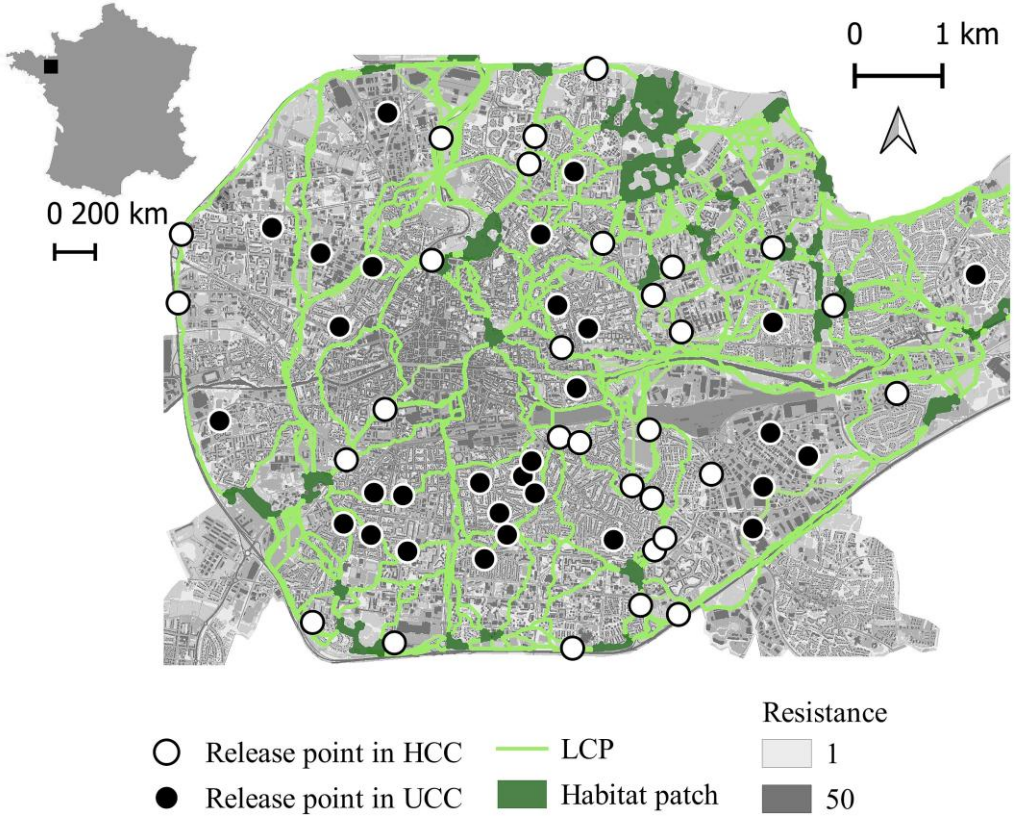
Neighbourhoods: Least cost path

- Calculates the lowest cost from a location to every other location and creates a distance surface in raster form.
- A least cost path represents the most efficient route through the landscape.
  e.g. wildlife corridors in fragmented habitat (foraging, range), pipeline or road planning, rescue operations, etc.



Hedgehog movement patterns in urban landscapes

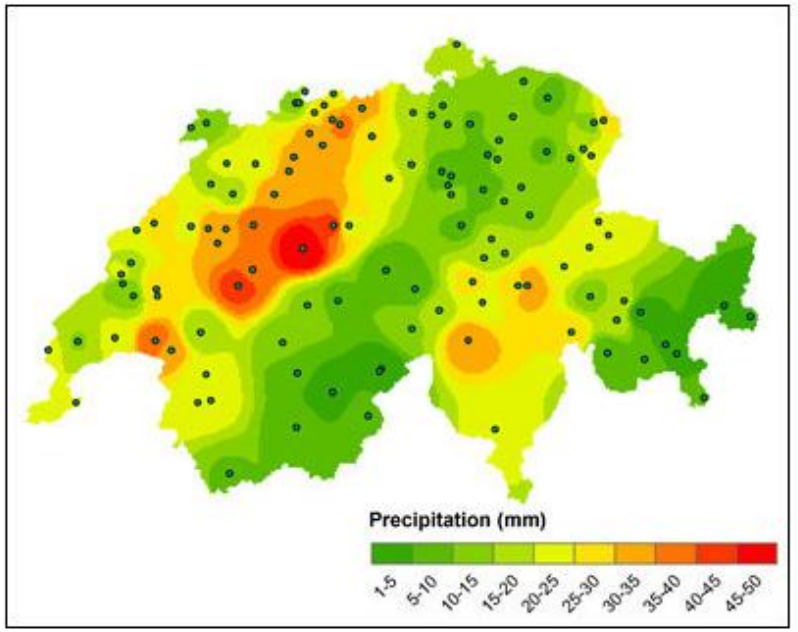
- Different hedgehog movement patterns depend upon the degree of predicted connectivity in urban landscapes.
- In highly connecting contexts (HCC), hedgehogs followed predicted low cost path (LCP) corridors, whereas in unconnected contexts (UCC) movements were random.



Balbi et al (2019)
https://doi.org/10.1016/j.jenvman.2019.04.124

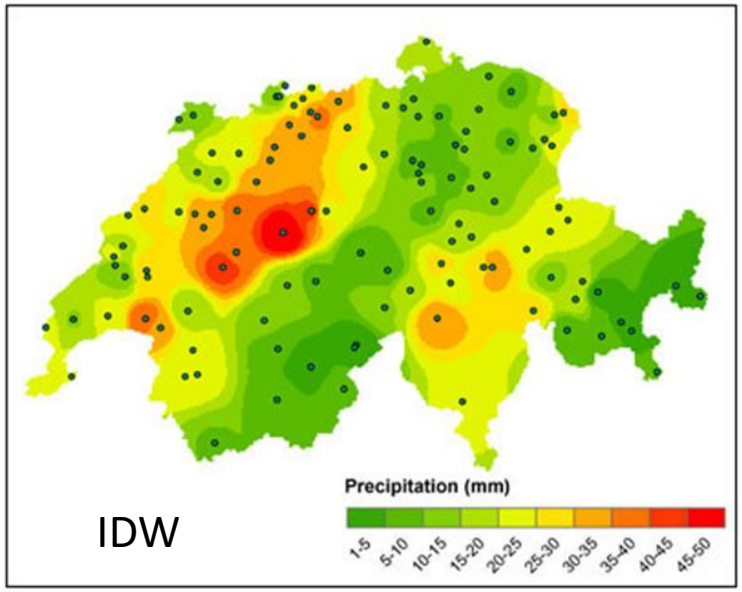
Spatial interpolation

- Estimating values of an environmental variable at locations whose values are unknown based on a sample of locations with known values.
- Apply an interpolation method to a set of points with known values to create a continuous surface.

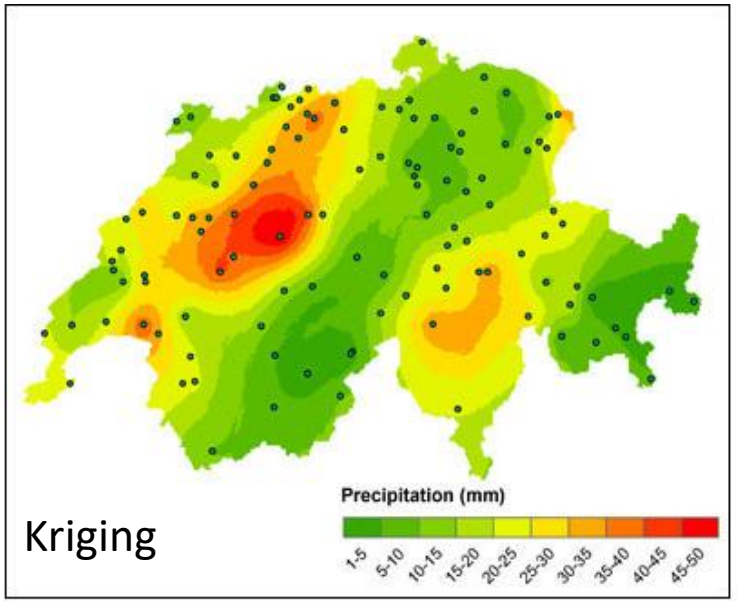


Spatial interpolation methods

- Environmental data are often collected as discrete observations at points.
- Two common methods: Inverse Distance Weights (IDW) = deterministic and Kriging = geostatistical.

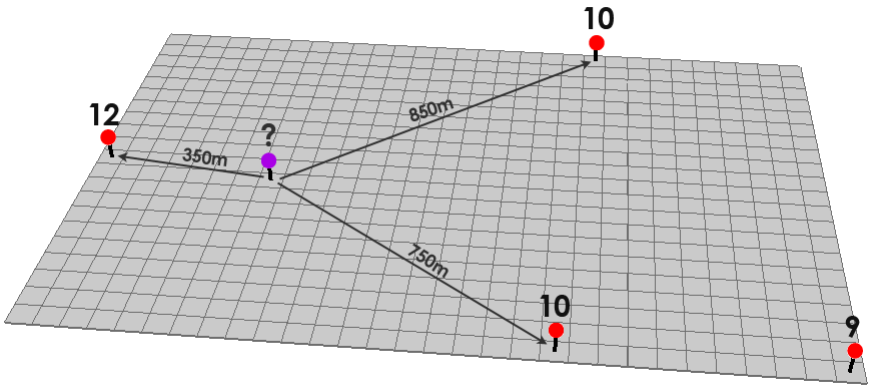


IDW



Kriging

Inverse Distance Weights (IDW)

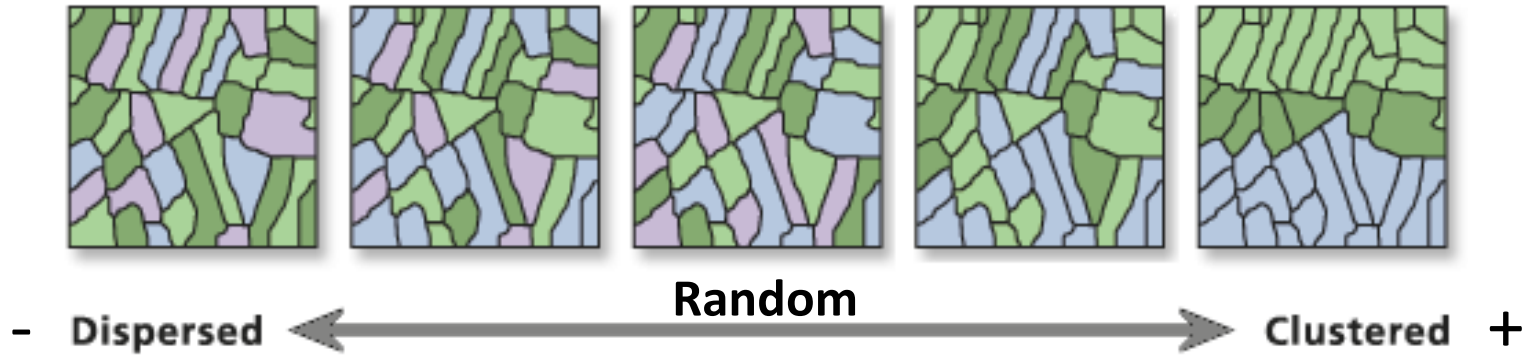


$$z_p = \frac{\sum_{i=1}^{n} \left(\frac{z_i}{d_i^p} \right)}{\sum_{i=1}^{n} \left(\frac{1}{d_i^p} \right)}$$

z_p is interpolated value, z_i is known value at points; d_i = distance of points from z_p ; p = power term.

- IDW estimates a value of each location by taking the distance-weighted average of the values of sample points in its neighbourhood.
- Explicitly assumes that things closer to one another are more alike than those that further apart (Tobler's first law of geography).
- As *p* increases, the weights for distant points decrease rapidly. If the *p* value is very high, only the immediate surrounding points will influence the prediction.

Spatial autocorrelation



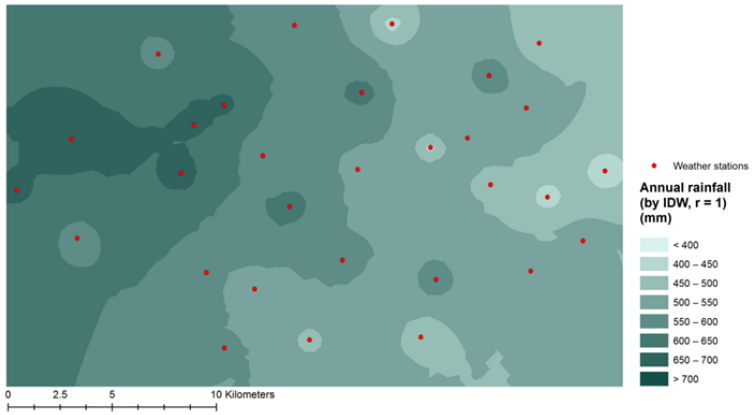
- Spatial autocorrelation is a statistical measure of Tobler's first law of geography "Everything is related to everything else, but near things are more related than distant things".
- **Positive** spatial autocorrelation is when similar values cluster together.
- **Negative** spatial autocorrelation is when dissimilar values cluster together (i.e. nearby locations are more different than expected by chance).
- Spatial autocorrelation is an underlying assumption of spatial interpolation.

Examples of IDW interpolation

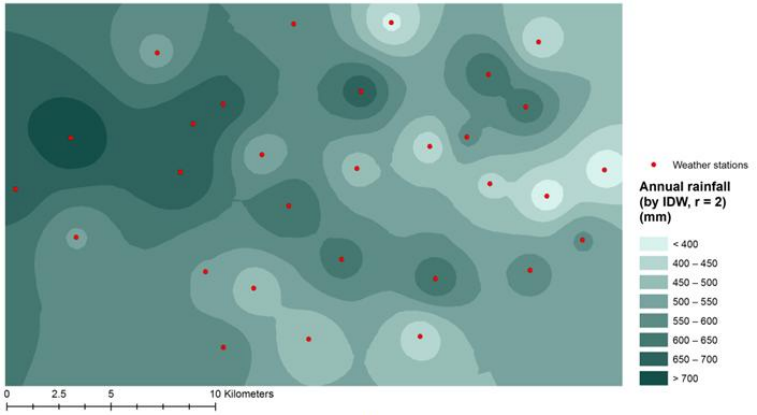
A power (p) of 1 smooths out the interpolated surface. Increases influence of more distant values.

$$Z_p = \frac{\sum_{i=1}^{n} \left(\frac{z_i}{d_i^p} \right)}{\sum_{i=1}^{n} \left(\frac{1}{d_i^p} \right)}$$

A power (p) of 2 increases the influence of close values.



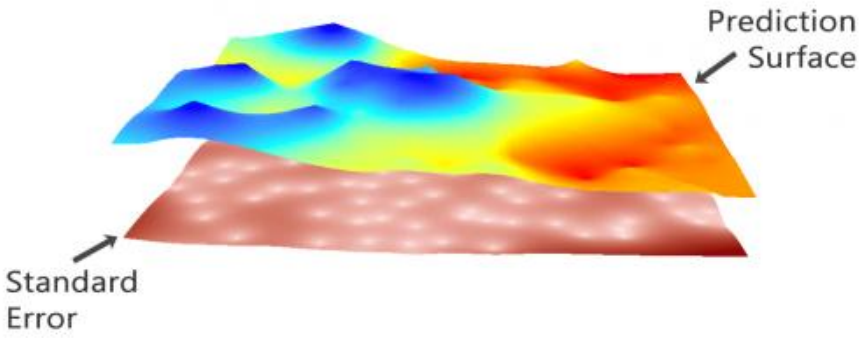
(a)



(b)

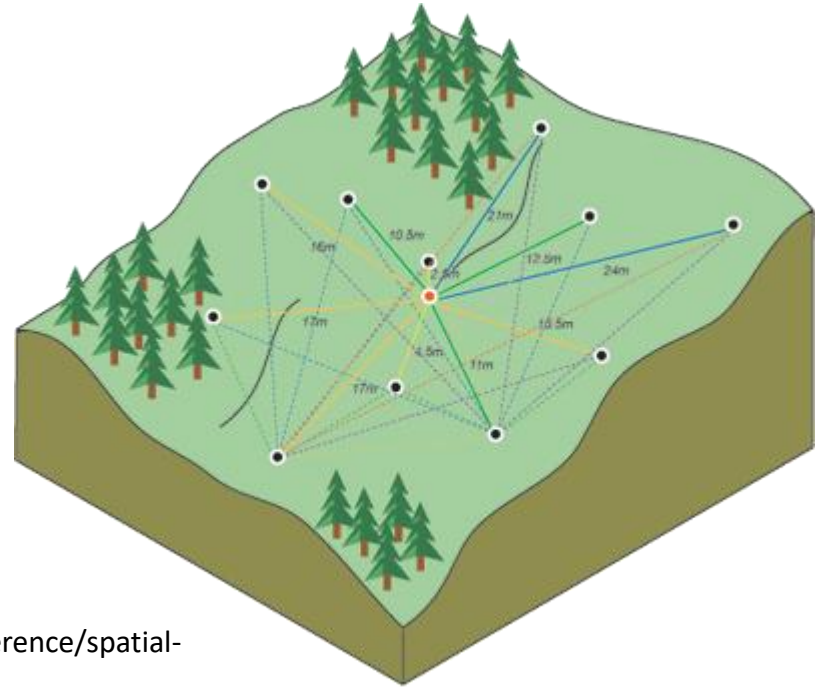
Kriging

- Kriging interpolates values and their probability at a specific location.
- Uses input data to build a mathematical function with a semivariogram, to create a prediction surface, and then validate the model with cross-validation.
- A geostatistical method that provides an optimal prediction surface AND a measure of confidence in how likely that prediction will be true.



Kriging – semivariance and distance

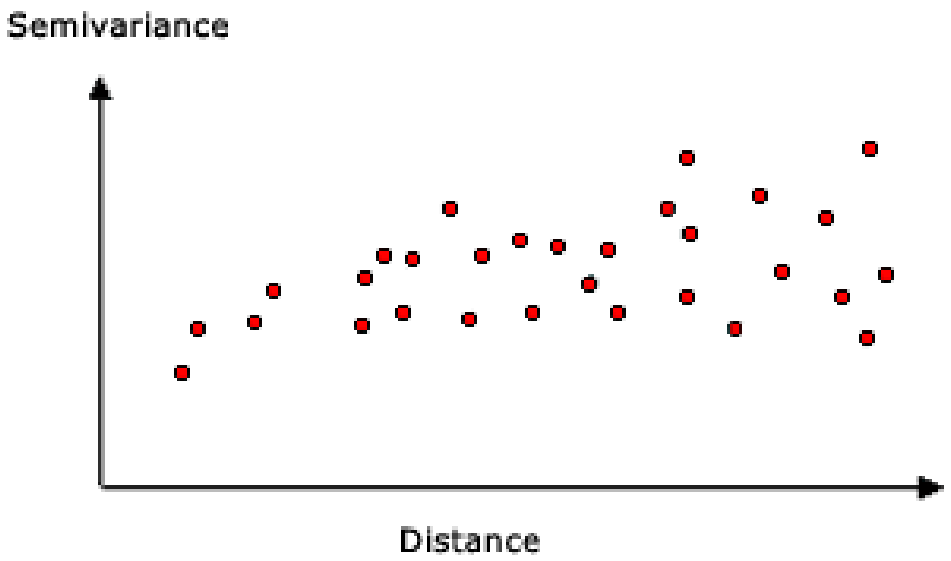
- Semivariogram(distance) = 0.5 * average(($value_i$ – $value_j$)^2)
- The formula involves calculating the difference squared between the values of the paired locations.
- The image shows the pairing of one point (the red point) with all other measured locations. This process continues for each measured point.



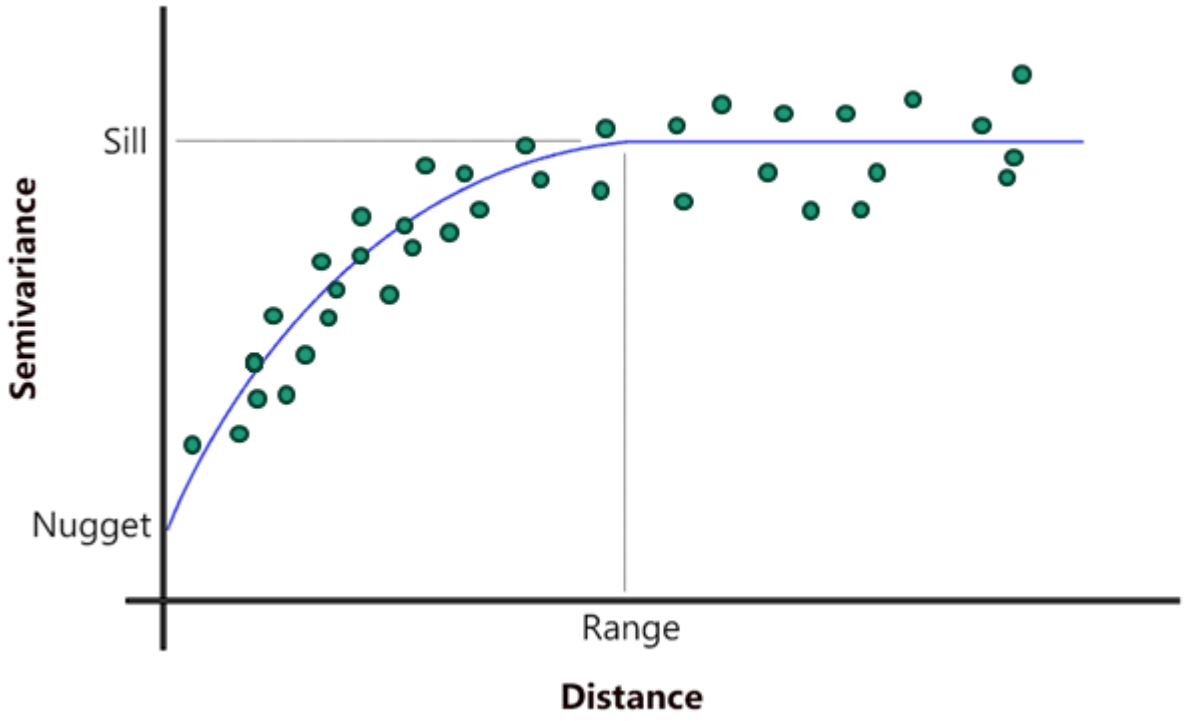
https://pro.arcgis.com/en/pro-app/latest/tool-reference/spatial-analyst/how-kriging-works.htm

Semivariogram

- The semivariogram is a graph of the averaged semivariogram values on the y-axis and the distance (or lag) on the x-axis.



Range, sill and nugget



The semivariogram defines the spatial structure that kriging uses to weight data points.

Ordinary kriging

- The semivariogram defines the spatial structure that kriging uses to weight data points.

- **Ordinary kriging** uses a linear combination of the observed data values, with weights, to make predictions of values at unmeasured locations.

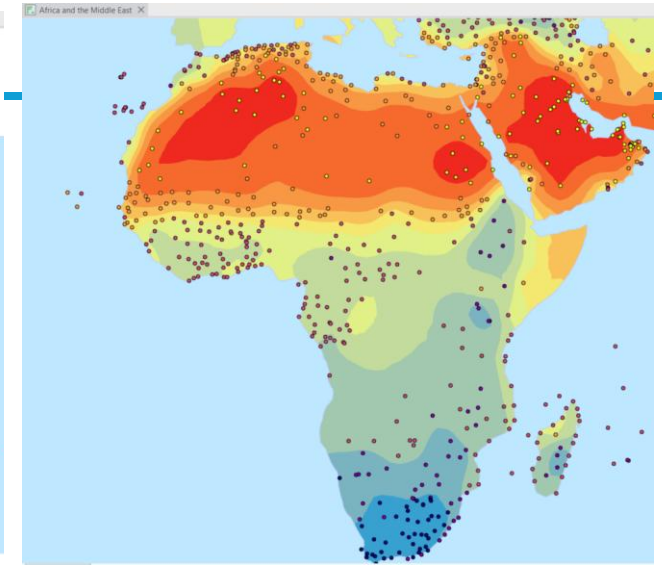
- **Simple kriging**

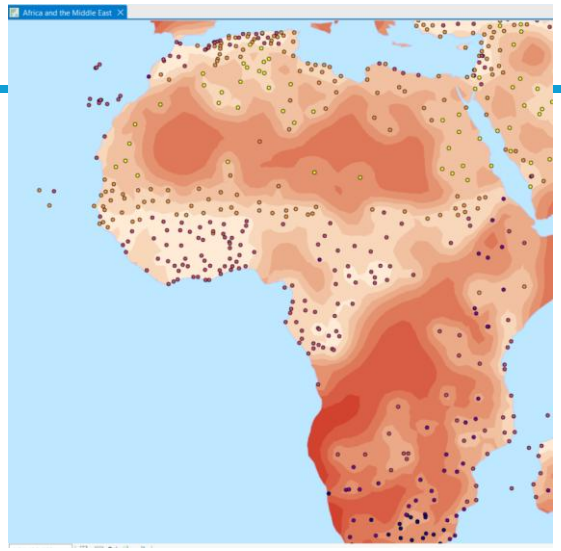
- **Universal kriging**

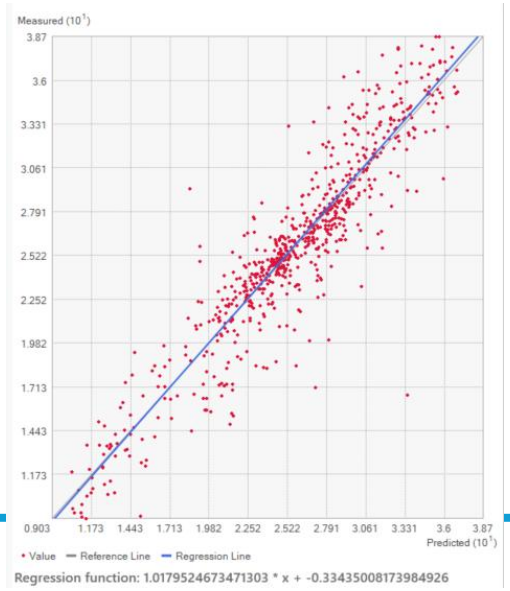
- **Co-kriging**

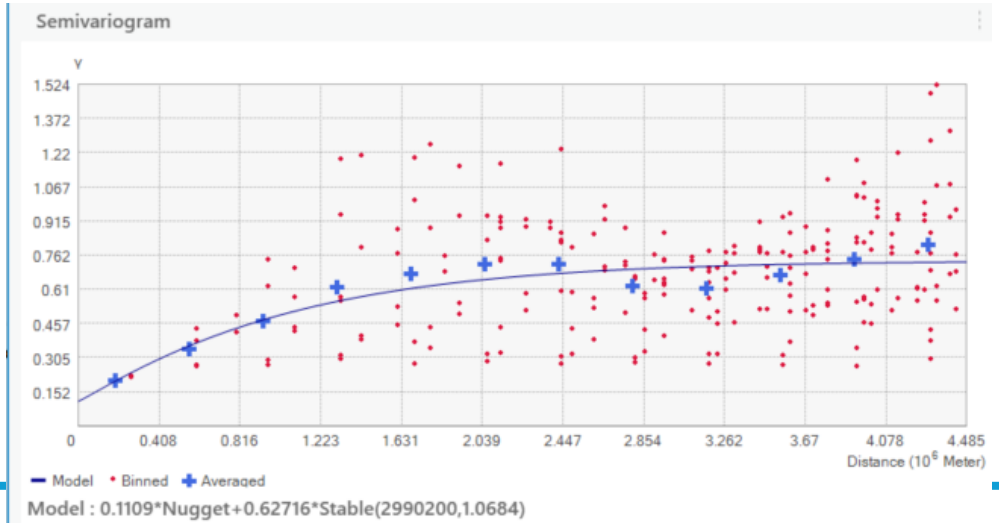
Kriging temperature variations









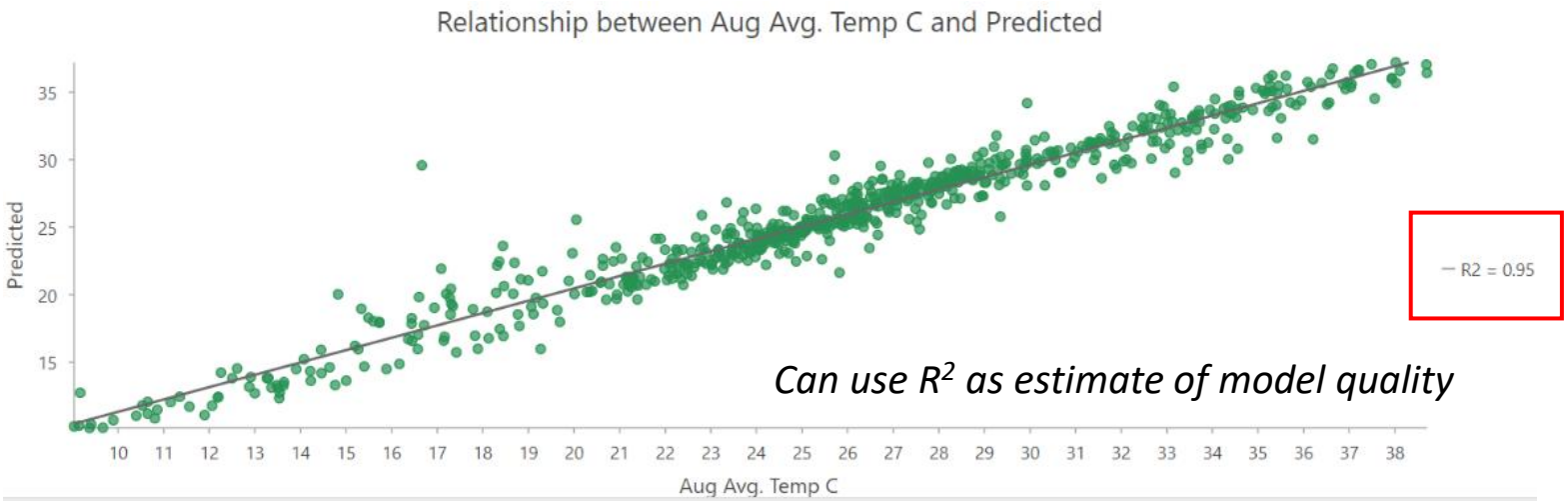


Kriging error analysis

Summary | Table

Count	704
Mean	-0.0610580007314659
Root-Mean-Square	2.29014197130688
Mean Standardized	-0.0109232984430551
Root-Mean-Square Standardized	0.846471403672981
Average Standard Error	2.89906646770786

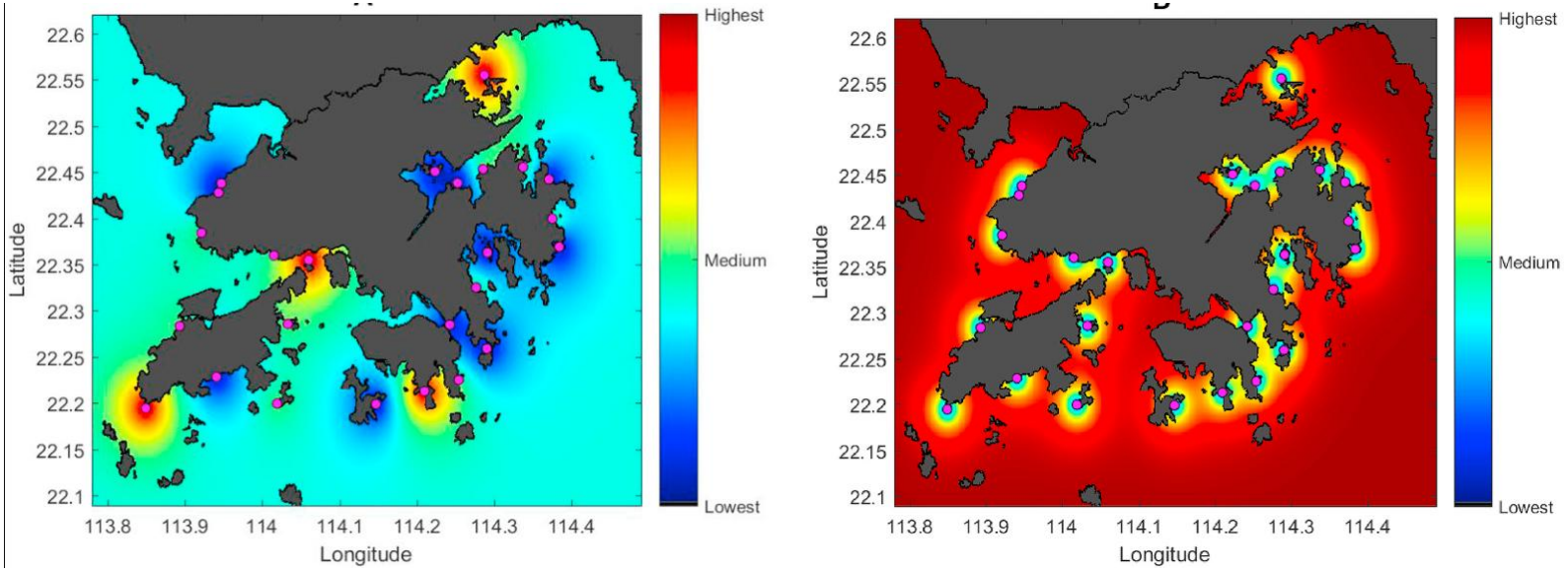
Count = number of known values.
Mean = average difference between known and predicted values (should be close to zero).
RMSE = prediction accuracy (should be as small as possible).
RMSS should be close to 1.



Can use R^2 as estimate of model quality

Kriging examples - plastics

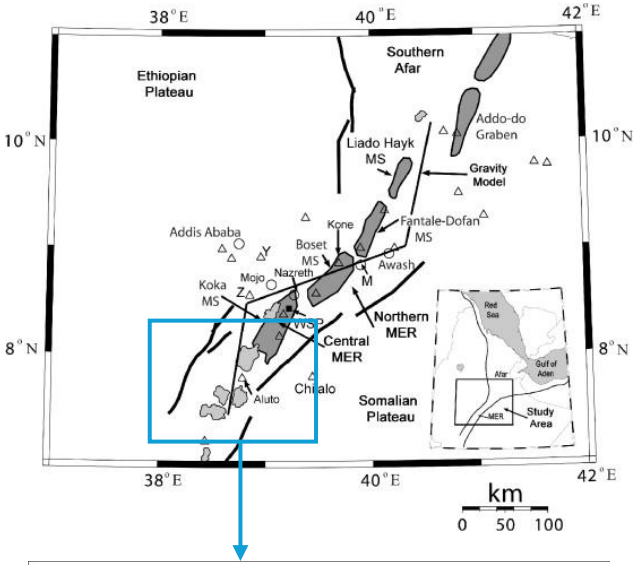
Coleby and Grist (2019) used a semivariograms and kriging to map the distribution of marine plastic pollution in Hong Kong.

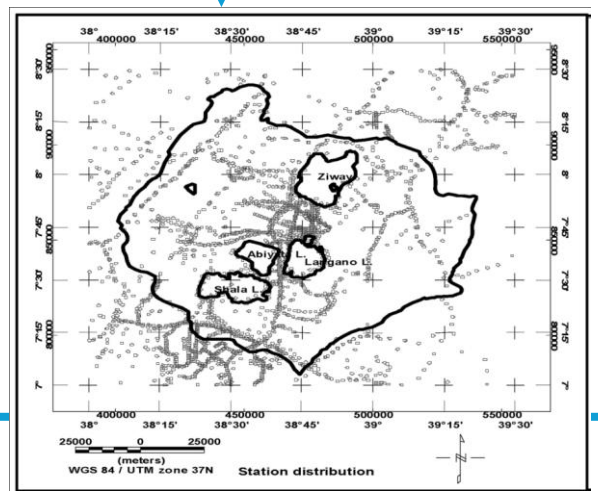


kriging variance

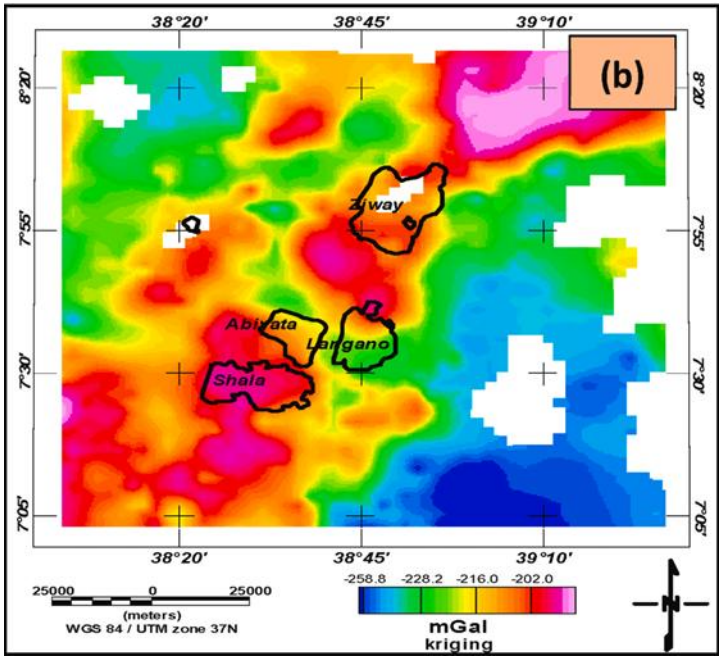
https://doi.org/10.1016/j.ocecoaman.2018.12.021

Kriging examples – crustal thickness





Kebede et al.(2024) used kriging to obtain a Bouguer anomaly map of the Main Ethiopian Rift (MER)



https://doi.org/10.1016/j.ocecoaman.2018.12.021

Sources of error in interpolation

- Errors inherent in interpolation method

  measured in terms of root mean square error, RMSE or r^2

- Sample data uncertainty

  too few sample points, limited or clustered distributions and uncertainty about locations and/or values.

- Edge effects

  distortions of the interpolated values near the boundary of the study area due to the lack of sample data outside the area.

- Outliers

  outliers can significantly affect the accuracy of kriging predictions.

Homework and Resources

- Read Chapter 5 of course textbook: Spatial Analysis.
- Take the quiz to test your basic understanding (on Canvas).

Further resources

ESRI training Getting started with spatial analysis (components 1-3 only; 1hr 15 min).

https://www.esri.com/training/catalog/5bca58f8f77b99238f845e1c/getting-started-with-spatial-analysis/